

Other

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING
B.TECH DEGREE SECOND SEMESTER SECOND INTERNAL EXAMINATION
JULY 2024

23-200-0204B OBJECT ORIENTED PROGRAMMING IN C++
(S2-EC A)
(2023 Scheme)

Time: 2 Hours

Maximum marks: 30

Course outcomes:

1. To understand basic concepts of Object – Oriented Programming.
2. Describe object - oriented paradigm with concepts of streams, classes, functions, data and objects.
3. To implement object- oriented programming constructs like encapsulation, inheritance and polymorphism.
4. To understand dynamic memory management techniques using pointers, constructors, destructors etc.
5. To identify classes including data, methods and the relationship among classes from a given problem statement and solve the problem using object -oriented constructs in C++

Part A

(Answer all questions)

Marks (5×2=10)

I.

- 1) Differentiate private and protected inheritance. (CO3, L2, PO-1,2,3,12)
- 2) With a suitable program describe hybrid inheritance. (CO3, L3, PO-1,2,3,12)
- 3) Write a short note on abstract classes. (CO2, L1, PO-1,2,12)
- 4) Explain file stream classes in C++. (CO2, L1, PO-1,2,12)
- 5) Explain file pointers in C++. (CO2, L2, PO-1,2,12)

Part B

Marks (2×10=20)

II.

1. (a) Create classes country, state and city. Display name and population of each using member function. Use multilevel inheritance. (CO3, L3, PO-1,2,3,12) (5)
- (b) WAP to calculate area of a square, triangle and circle using function overloading. (CO3, L3, PO-1,2,3,12) (5)

OR

- 2.(a) WAP to define 3 classes A, B and C. Each class contains private data members. Derive class C from A and B by applying multiple inheritance. Read and display the data using constructors and destructors. (CO3, L3, PO-1,2,3,12) (5)

(b) WAP to overload `<` operator and display the smallest number out of two objects (CO3, L3, PO-1,2,3,12) (5)

3. (a) Describe generic programming using templates. WAP to show values of different data types using constructor and template (CO3, L3, PO-1,2,3,12) (5)

(b) WAP to read and display contents of a file using `eof()` function. (CO2, L3, PO-1,2,12) (5)

OR

4. (a) In detail explain exception handling with a suitable program. (CO2, L2, PO-1,2,12) (5)

(b) WAP to read a string as user input, write it to a file and display the string by reading it from file. Use `put()` and `get()` functions. (CO2, L3, PO-1,2,12) (5)

(L1-13%, L2-30%, L3-57%)

>>>><<<<